DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM
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ALGEBRA (Ph.D. Version)

Instructions to the student

a. Answer all six questions; each will be assigned a grade from 0 to 10.

b. Use a different booklet for each question. Write the problem number and your **code number** (**not** your name) on the outside of the booklet.

c. Keep scratch work on separate pages in the same booklet.

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1. Let G be a group acting on a set S. We say that the action is *doubly transitive* if, given $a, b, c, d \in S$ with $a \neq b$ and $c \neq d$, there exists $g \in G$ such that $ga = c$ and $gb = d$.
(a) Suppose G is a subgroup of S_n, the group of permutations of $S = \{1, 2, \ldots, n\}$, and G acts doubly transitively on S. Show that $n(n-1)$ divides the order of G. (*Hint:* G acts on the set of $n(n-1)$ ordered pairs $(a, b) \in S \times S)$ with $a \neq b$.)
(b) Let A_5 be the group of even permutations of the set $\{1, 2, 3, 4, 5\}$ and suppose G (acting on this set) is a doubly transitive subgroup of A_5 (therefore a doubly transitive subgroup of S_5). Show that $G = A_5$. (You may use the fact that A_5 is simple.)

2. Let A be an $n \times n$ matrix with complex entries. Define

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots$$

(you may assume that the series converges). Show that $\det(e^A) = e^{\mathrm{Trace}(A)}$.

3. Let R be a commutative ring with 1. Recall that an element $a \in R$ is *nilpotent* if $a^k = 0$ for some integer $k \geq 1$.
(a) Let $a \in R$. Let I be an ideal of R such that $I \cap \{1, a, a^2, \ldots\}$ is empty, and suppose I is maximal with respect to this property. Show that I is a prime ideal.
(b) Show that if $a \in R$ is not nilpotent, then there is a prime ideal that does not contain a. (*Hint:* Zorn's Lemma.) Indicate where you use the assumption that a is not nilpotent.
(c) Show that the intersection of all prime ideals of R is the set of nilpotent elements of R.

4. Let R be an integral domain. Recall that an R-module is projective if it is a direct summand of a free module. An R-module I is injective if, whenever there are R-modules A and B with $A \subseteq B$ and an R-module homomorphism $f : A \to I$, then there is an R-module homomorphism $g : B \to I$ such that $g = f$ on A.
(a) Let F be the field of fractions of R. Show that if F is a projective R-module then $R = F$.
(b) Suppose R is an injective R-module. Show that R is a field.

5. Let p be an odd prime and let $K/\mathbb{Q}$ be a Galois extension of degree $2p$. Let $G = \mathrm{Gal}(K/\mathbb{Q})$.
(a) Show that there is a unique field $F \subset K$ such that $[F : \mathbb{Q}] = 2$.
(b) Show that there is a unique field $L \subset K$ with $[L : \mathbb{Q}] = p$ if and only if G is abelian.
(c) Show that there is an irreducible polynomial in $\mathbb{Q}[X]$ of degree p whose splitting field is K if and only if G is nonabelian.

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6. (a) Let $\{e_1, e_2, e_3, e_4\}$ be the standard basis of $\mathbb{C}^4$. The group A_4 of even permutations acts on $\mathbb{C}^4$ by permuting the basis elements: if $\sigma \in S_4$, then $e_i \mapsto e_{\sigma(i)}$. This yields a representation ρ. Show that ρ is the sum of the trivial representation and an irreducible representation of dimension 3.
(b) Find the dimensions of the irreducible representations of A_4.